DataBase Lab 04 (POST-Lab Task)

Muhammad Ghulam Mujtaba Qureshi

24k-0535

Task 11

Welcome Page | HR

Worksheet | Query Builder

```
SELECT first_name || ' ' || last_name AS emp_name,
       salary,
       ROUND((salary / SUM(salary) OVER()) * 100, 2) AS salary_percentage
FROM employees;
```

Query Result

SQL | Fetched 50 rows in 0.003 seconds

	EMP_NAME	SALARY	SALARY_PERCENTAGE
1	Steven King	24000	3.47
2	Neena Kochhar	17000	2.46
3	Lex De Haan	17000	2.46
4	Alexander Hunold	9000	1.3
5	Bruce Ernst	6000	0.87
6	David Austin	4800	0.69
7	Valli Pataballa	4800	0.69
8	Diana Lorentz	4200	0.61
9	Nancy Greenberg	12008	1.74
10	Daniel Faviet	9000	1.3
11	John Chen	8200	1.19
12	Ismael Sciarra	7700	1.11
13	Jose Manuel Urman	7800	1.13
14	Luis Popp	6900	1
15	Den Raphaely	11000	1.59
16	Alexander Khoo	3100	0.45
17	Shelli Baida	2900	0.42
18	Sigal Tobias	2800	0.4
19	Guy Himuro	2600	0.38
20	Karen Colmenares	2500	0.36
21	Matthew Weiss	8000	1.16
22	Adam Fripp	8200	1.19
23	Payam Kaufling	7900	1.14
24	Shanta Vollman	6500	0.94
25	Kevin Mourgos	5800	0.84
26	Julia Nayer	3200	0.46
27	Irene Mikkilineni	2700	0.39
28	James Landry	2400	0.35
29	Steven Markle	2200	0.32
30	Laura Bissot	3300	0.48
31	Mozhe Atkinson	2800	0.4
32	James Marlow	2500	0.36

Task 12

Welcome Page | HR

Worksheet | Query Builder

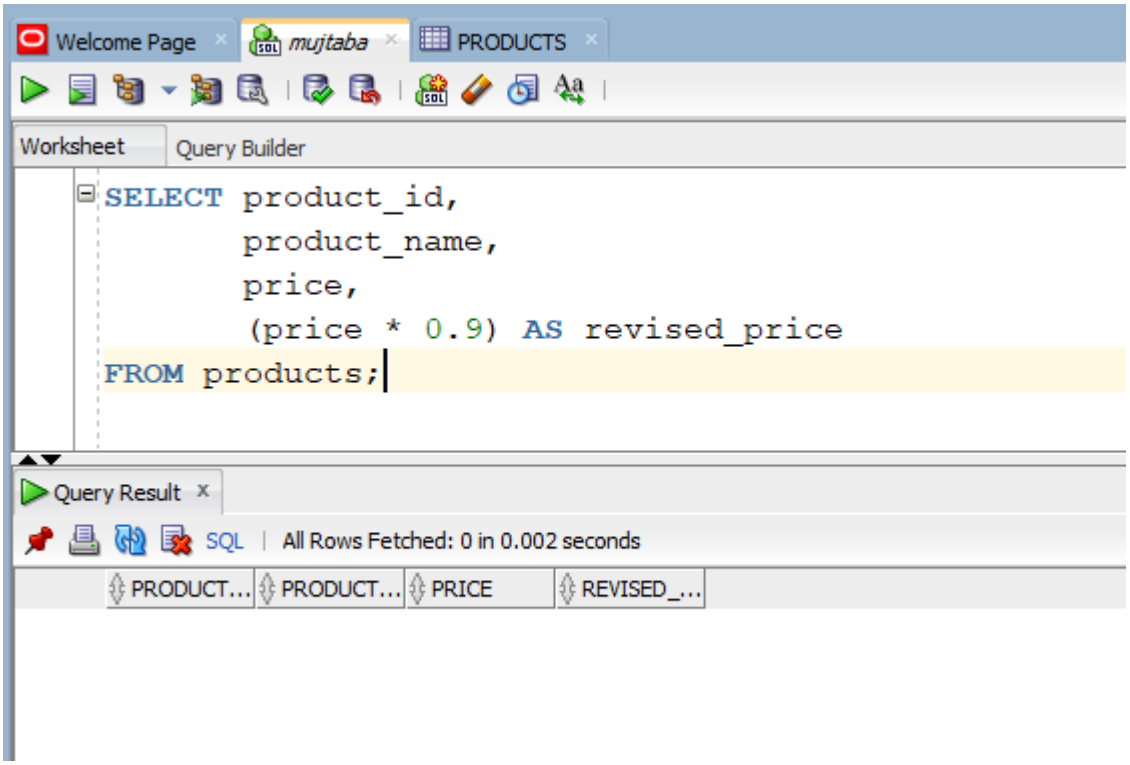
```
SELECT UPPER(SUBSTR(department_name, 1, 3)) AS dept_code
FROM departments;
```

Query Result

SQL | All Rows Fetched: 27 in 0.002 seconds

	DEPT_CODE
1	ADM
2	MAR
3	PUR
4	HUM
5	SHI
6	IT
7	PUB
8	SAL
9	EXE
10	FIN
11	ACC
12	TRE
13	COR
14	CON
15	SHA
16	BEN
17	MAN
18	CON
19	CON
20	OPE
21	IT
22	NOC
23	IT
24	GOV
25	RET
26	REC
27	PAY

Task 13


Welcome Page
mujtaba
PRODUCTS
Worksheet
Query Builder
SELECT product_id,
product_name,
price,
(price * 0.9) AS revised_price
FROM products;
Query Result
SQL
All Rows Fetched: 0 in 0.002 seconds
PRODUCT...
PRODUCT...
PRICE
REVISED_...

Task 14

Welcome Page | HR

Worksheet | Query Builder

```sql
SELECT d.department_name,
       ROUND(AVG(e.salary), 2) AS avg_salary
FROM employees e
JOIN departments d ON e.department_id = d.department_id
GROUP BY d.department_name;
```

Query Result

SQL | All Rows Fetched: 11 in 0.024 seconds

	DEPARTMENT_NAME	AVG_SALARY
1	Administration	4400
2	Accounting	10154
3	Purchasing	4150
4	Human Resources	6500
5	IT	5760
6	Public Relations	10000
7	Executive	19333.33
8	Shipping	3475.56
9	Sales	8955.88
10	Finance	8601.33
11	Marketing	9500

Task 15

Welcome Page | HR

Worksheet | Query Builder

```
SELECT first_name,last_name,
       NVL(TO_CHAR(commission_pct), 'No Commission') AS commission
FROM employees;
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.002 seconds

	FIRST_NAME	LAST_NAME	COMMISSION
1	Steven	King	No Commission
2	Neena	Kochhar	No Commission
3	Lex	De Haan	No Commission
4	Alexander	Hunold	No Commission
5	Bruce	Ernst	No Commission
6	David	Austin	No Commission
7	Valli	Pataballa	No Commission
8	Diana	Lorentz	No Commission
9	Nancy	Greenberg	No Commission
10	Daniel	Faviet	No Commission
11	John	Chen	No Commission
12	Ismael	Sciarra	No Commission
13	Jose Manuel	Urman	No Commission
14	Luis	Popp	No Commission
15	Den	Raphaely	No Commission
16	Alexander	Khoo	No Commission
17	Shelli	Baida	No Commission
18	Sigal	Tobias	No Commission
19	Guy	Himuro	No Commission
20	Karen	Colmenares	No Commission
21	Matthew	Weiss	No Commission
22	Adam	Fripp	No Commission
23	Payam	Kaufling	No Commission
24	Shanta	Vollman	No Commission
25	Kevin	Mourgos	No Commission
26	Julia	Nayer	No Commission
27	Irene	Mikkilineni	No Commission
28	James	Landry	No Commission
29	Steven	Markle	No Commission
30	Laura	Bissot	No Commission
31	Mozhe	Atkinson	No Commission
32	James	Marlow	No Commission

Task 16

Welcome Page | HR

Worksheet | Query Builder

```
SELECT UPPER(first_name || ' ' || last_name) AS emp_name_upper,
       SUBSTR(job_title, 1, 4) AS job_abbrev,
       TO_CHAR(hire_date, 'DD-MON-YYYY') AS formatted_hire_date
FROM employees e
JOIN jobs j ON e.job_id = j.job_id;
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.001 seconds

	EMP_NAME_UPPER	JOB_ABBREV	FORMATTED_HIRE_DATE
1	WILLIAM GIETZ	Publ	07-JUN-2002
2	SHELLEY HIGGINS	Acco	07-JUN-2002
3	JENNIFER WHALEN	Admi	17-SEP-2003
4	STEVEN KING	Pres	17-JUN-2003
5	LEX DE HAAN	Admi	13-JAN-2001
6	NEENA KOCHHAR	Admi	21-SEP-2005
7	JOHN CHEN	Acco	28-SEP-2005
8	DANIEL FAVIET	Acco	16-AUG-2002
9	LUIS POPP	Acco	07-DEC-2007
10	ISMAEL SCIARRA	Acco	30-SEP-2005
11	JOSE MANUEL URMAN	Acco	07-MAR-2006
12	NANCY GREENBERG	Fina	17-AUG-2002
13	SUSAN MAVRIS	Huma	07-JUN-2002
14	ALEXANDER HUNOLD	Prog	03-JAN-2006
15	BRUCE ERNST	Prog	21-MAY-2007
16	DIANA LORENTZ	Prog	07-FEB-2007
17	VALLI PATABALLA	Prog	05-FEB-2006
18	DAVID AUSTIN	Prog	25-JUN-2005
19	MICHAEL HARTSTEIN	Mark	17-FEB-2004
20	PAT FAY	Mark	17-AUG-2005
21	HERMANN BAER	Publ	07-JUN-2002
22	SHELLI BAIDA	Purc	24-DEC-2005
23	ALEXANDER KHOO	Purc	18-MAY-2003
24	SIGAL TOBIAS	Purc	24-JUL-2005
25	GUY HIMURO	Purc	15-NOV-2006
26	KAREN COLMENARES	Purc	10-AUG-2007
27	DEN RAPHAELY	Purc	07-DEC-2002
28	GERALD CAMBRAULT	Sale	15-OCT-2007
29	ALBERTO ERRAZURIZ	Sale	10-MAR-2005

Task 17

Welcome Page | HR

Worksheet | Query Builder

```sql
SELECT first_name || ' ' || last_name AS emp_name,
       salary,
       hire_date
FROM employees
WHERE (first_name LIKE '%n' OR first_name LIKE '%a%')
  AND salary < 80000
  AND hire_date BETWEEN TO_DATE('2012-01-01', 'YYYY-MM-DD')
                    AND TO_DATE('2022-12-31', 'YYYY-MM-DD');
```

Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.002 seconds

EMP_NAME	SALARY	HIRE_DATE

Task 18

Welcome Page | HR

Worksheet | Query Builder

```
SELECT first_name || ' ' || last_name AS emp_name,
       salary,
       (salary * 12) AS annual_income
FROM employees;
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.002 seconds

	EMP_NAME	SALARY	ANNUAL_INCOME
1	Steven King	24000	288000
2	Neena Kochhar	17000	204000
3	Lex De Haan	17000	204000
4	Alexander Hunold	9000	108000
5	Bruce Ernst	6000	72000
6	David Austin	4800	57600
7	Valli Pataballa	4800	57600
8	Diana Lorentz	4200	50400
9	Nancy Greenberg	12008	144096
10	Daniel Faviet	9000	108000
11	John Chen	8200	98400
12	Ismael Sciarra	7700	92400
13	Jose Manuel Urman	7800	93600
14	Luis Popp	6900	82800
15	Den Raphaely	11000	132000
16	Alexander Khoo	3100	37200
17	Shelli Baida	2900	34800
18	Sigal Tobias	2800	33600
19	Guy Himuro	2600	31200
20	Karen Colmenares	2500	30000
21	Matthew Weiss	8000	96000
22	Adam Fripp	8200	98400
23	Payam Kaufling	7900	94800
24	Shanta Vollman	6500	78000
25	Kevin Mourgos	5800	69600
26	Julia Nayer	3200	38400
27	Irene Mikkilineni	2700	32400
28	James Landry	2400	28800
29	Steven Markle	2200	26400
30	Laura Bissot	3300	39600
31	Mozhe Atkinson	2800	33600
32	James Marlow	2500	30000

Task 19

Welcome Page | HR

Worksheet | Query Builder

```
SELECT first_name || ' ' || last_name AS emp_name,
       salary,
       hire_date
FROM employees
WHERE (first_name LIKE 'A%' OR first_name LIKE 'M%')
  AND salary > 70000
  AND hire_date BETWEEN TO_DATE('2010-01-01', 'YYYY-MM-DD')
                    AND TO_DATE('2020-12-31', 'YYYY-MM-DD');
```

Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.002 seconds

EMP_NAME	SALARY	HIRE_DATE

Task 20

Welcome Page | HR

Worksheet | Query Builder

```
SELECT first_name || ' ' || last_name AS emp_name,
       d.department_name,
       FLOOR(MONTHS_BETWEEN(SYSDATE, hire_date)/12) AS tenure_years
FROM employees e
JOIN departments d ON e.department_id = d.department_id
WHERE FLOOR(MONTHS_BETWEEN(SYSDATE, hire_date)/12) > 5
  AND salary > 60000;
```

Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.003 seconds

EMP_NAME	DEPARTM...	TENURE_...